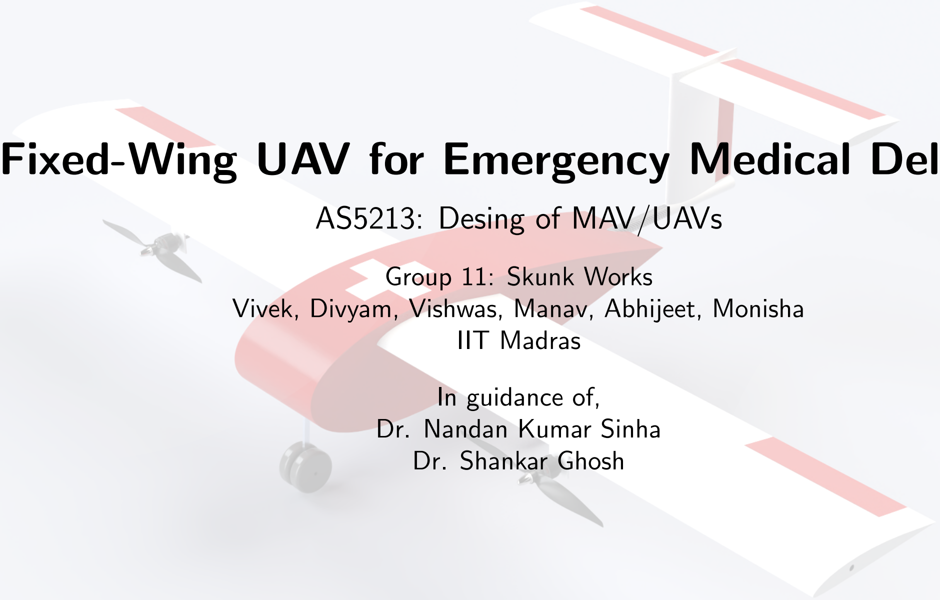

Fixed-Wing UAV for Emergency Medical Delivery

AS5213: Desing of MAV/UAVs

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IIT Madras

In guidance of,
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Presentation Roadmap

Mission and concept

- Mission profile and requirements
- Design methodology
- Aircraft configuration

Technical data

- Powerplant and energy budget
- Aerodynamic parameters
- Mass properties and inertia

Verification

- Performance envelope
- Stability and eigenvalues
- Requirements compliance

Goal

Show that the UAV meets mission range, payload, STOL, and stability targets with a practical conceptual design.

Mission Requirements

Mission objective

A fixed-wing UAV to deliver antivenom and blood products to remote locations with a droppable payload and return capability.

Top-level requirements

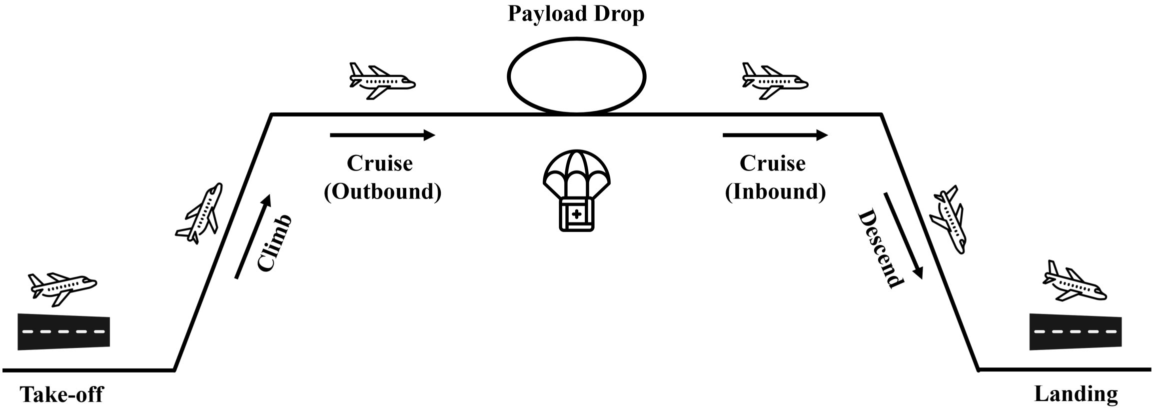
- Wingspan limited to 2.00 m
- Payload mass around 1 kg
- Operational radius 40–50 km
- Total range 80–100 km
- Cruise speed 18–25 m/s
- Takeoff distance below 150 m

Mission phases

1. Takeoff and climb
2. Outbound cruise
3. Payload drop
4. Return cruise
5. Descent and landing

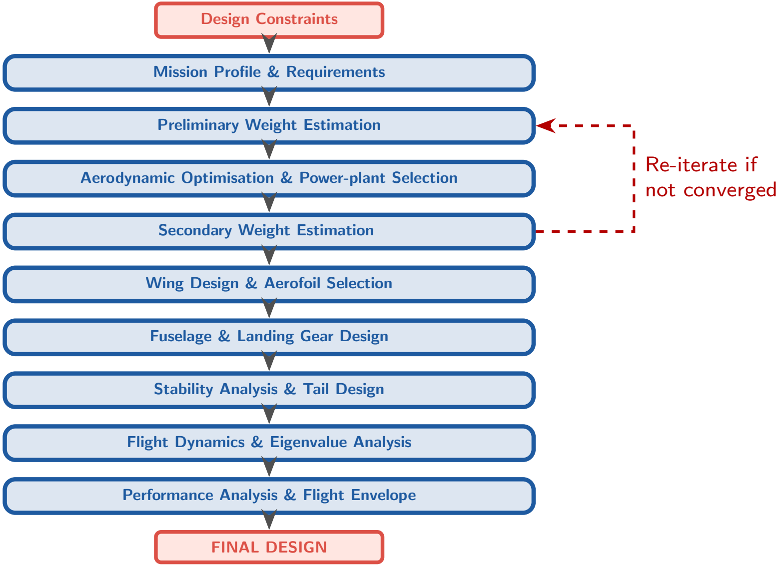
Design driver: range and payload have to remain compatible with STOL operation.

Mission Profile



Mission timeline used to size propulsion, energy storage, and payload-drop events.

Design Methodology Flowchart



Technical Data I: Powerplant

Propulsion and energy system

Motor	T-Motor AM480 900KV
Configuration	Dual tractor motors
Maximum power	450 W at 80% throttle
Supply voltage	11.1 V (3S LiPo)
Peak current	26 A
ESC	Hobbywing Skywalker 50A V2
Propeller	APC 12×6E

Power budget

Climb power: 361.46 W

Avionics: 6.61 W

Battery efficiency: 85%

Motor efficiency: 82%

Component imagery

T-Motor AM480 900KV

APC 12×6E

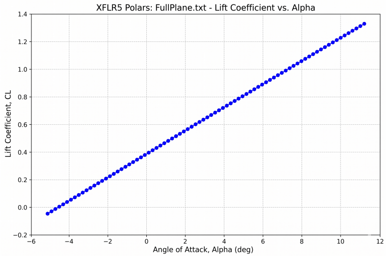
Installation note

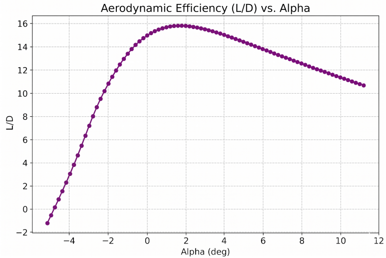
Dual-tractor wing-mounted propulsion with the battery placed to help CG control.

Technical Data II: Aerodynamics

Wing and drag data

Airfoil	Selig S7055
Planform	Rectangular, high-wing
Aspect ratio	7.53
Wing area	$0.5307\,\mathrm{m}^2$
Wingspan	$2.00\,\mathrm{m}$
Mean chord	$0.2653\,\mathrm{m}$
Zero-lift drag	$C_{D_0} = 0.02873$
Oswald efficiency	$e = 0.85$
Max L/D	12.55





Technical Data III: Mass Properties

Mass breakdown

MTOW	6.59 kg
Empty weight	3.29 kg
Battery	1.53 kg
Payload	1.20 kg
Avionics	0.60 kg
Static margin	$\approx$ 13% MAC

Empennage and controls

Horizontal tail area	$0.133\,\mathrm{m}^2$
Horizontal tail span	0.773 m
Tail arm	0.70 m
Tail airfoil	NACA 0012
Tail setting angle	-0.31°
Vertical tail area	$0.080\,\mathrm{m}^2$

Geometry

Fuselage length	1.10 m
Fuselage width	0.20 m
Tail boom length	0.448 m
CG location	264 mm from nose
Neutral point	296 mm

Design note

The payload bay is placed near the CG and supports droppable payload release without destabilizing the aircraft.

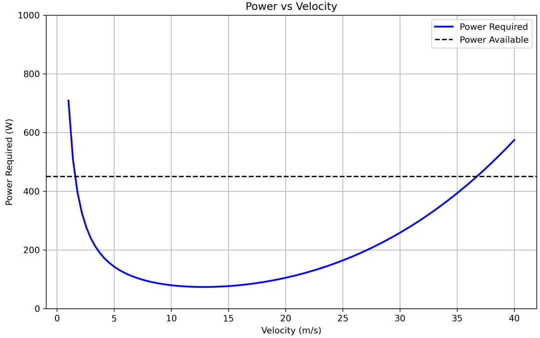
Performance I: Power Required

Performance summary

- Cruise speed: 18 m/s
- Maximum range: 83.15 km
- Endurance: 2 h 5 min
- Maximum speed: 35 m/s

Power margin

The required power remains below available power across the practical cruise band.



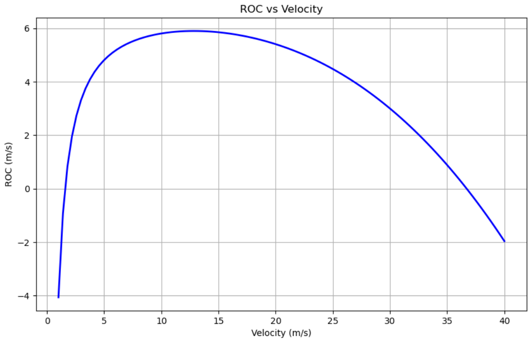
Performance II: Rate of Climb

Climb performance

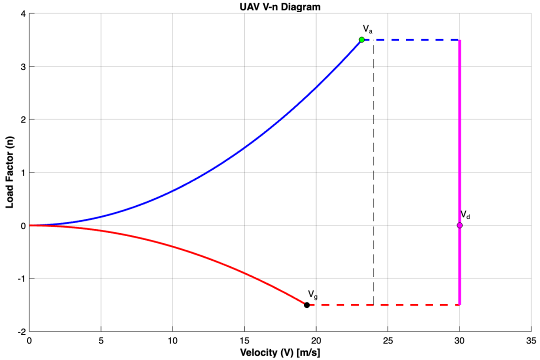
- Rate of climb: upto 6 m/s
- Optimal climb speed: 16 m/s
- Climb angle: 5°
- Takeoff distance: ∼150 m

Takeaway

Climb performance is adequate for the mission and remains inside the available power margin.



Performance III: Load Factor and Takeoff Envelope



Positive maneuver envelope before payload release.

Takeoff and landing

Takeoff distance	$< 150\,\text{m}$
Liftoff speed	$13.6\,\text{m/s}$
Landing distance	$< 120\,\text{m}$
Approach speed	$15\,\text{m/s}$
Positive limit load	3.5
Negative limit load	-1.5

Operational takeaway

The aircraft satisfies the STOL requirement while retaining a practical speed margin for climb, cruise, and landing.

Stability I: Static Derivatives

Longitudinal stability

C_{m_α}	$-0.487\ \mathrm{rad}^{-1}$
C_{m_0}	$+0.0125$
C_{m_q}	$-14.94\ \mathrm{rad}^{-1}$
C_{L_q}	$+5.66\ \mathrm{rad}^{-1}$
C_{D_α}	$+0.104\ \mathrm{rad}^{-1}$

Lateral-directional stability

C_{n_β}	$+0.0838\ \mathrm{rad}^{-1}$
C_{l_β}	$-0.0111\ \mathrm{rad}^{-1}$
C_{l_p}	$-0.659\ \mathrm{rad}^{-1}$
C_{n_r}	negative, stabilizing
Roll damping	strong

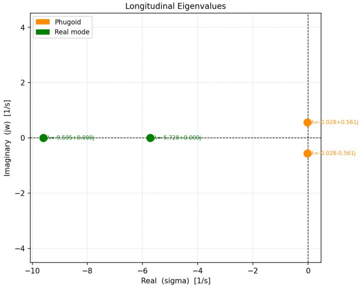
Interpretation

Negative C_{m_α} indicates static longitudinal stability, and the CG remains ahead of the neutral point.

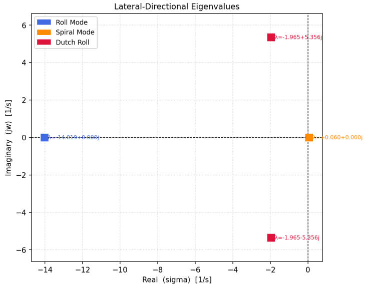
Interpretation

The sign pattern is consistent with weathercock stability, roll damping, and a conventional tail layout.

Stability II: Eigenvalue Analysis



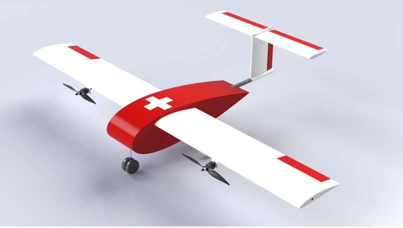
Longitudinal modes: phugoid and real mode.



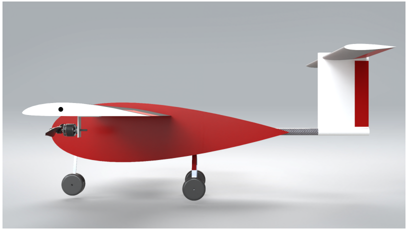
Lateral-directional modes: roll, spiral, and Dutch roll.

Mode	Key result	Interpretation
Phugoid	lightly damped	expected for small UAV
Roll mode	fast decay	strong roll stabilization
Dutch roll	stable complex pair	acceptable damping
Spiral mode	slightly unstable / slow	monitor in trim and control tuning

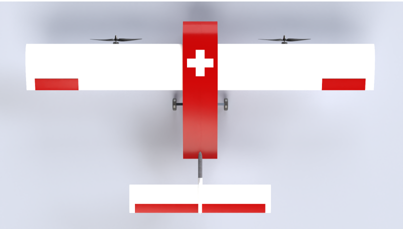
CAD Views: Orthographic & Isometric



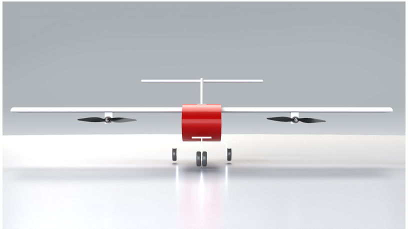
Isometric View



Side View

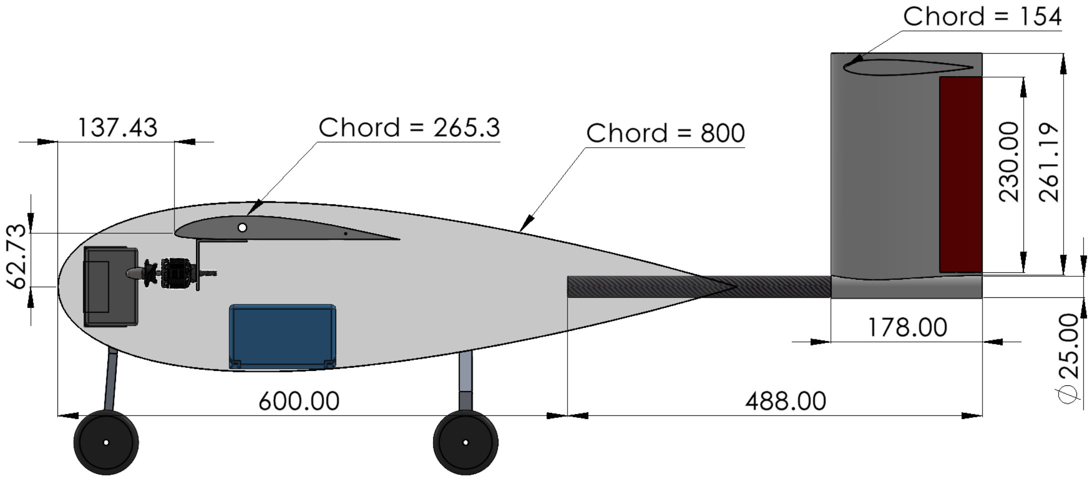


Top View



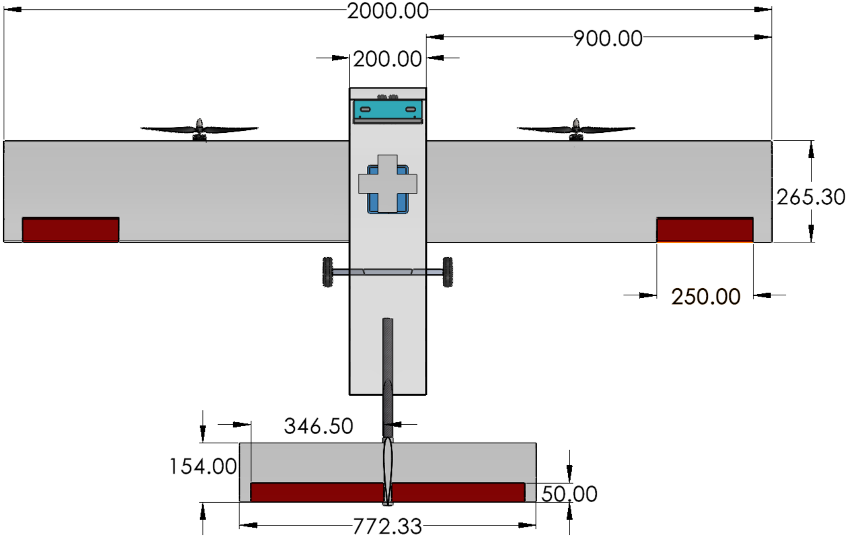
Front View

UAV Dimensional Specifications



Overall Dimensions from Side view

UAV Dimensional Specifications



Overall Dimensions from Top view

Requirements Compliance Summary

Requirement	Target	Achieved	Status
Maximum wingspan	2.00 m	2.00 m	✓
Payload capacity	1–2 kg	1.20 kg	✓
Operational radius	40–50 km	41.6 km	✓
Total range	80–100 km	83.15 km	✓
Endurance	1–2 h	2 h 5 min	✓
Cruise speed	18–25 m/s	18–22 m/s	✓
Takeoff distance	$<$ 150 m	$<$ 150 m	✓
Static stability	positive margin	$\approx$ 13% MAC	✓

Overall conclusion: the conceptual UAV meets the mission, range, payload, STOL, and stability goals.

Thank you

Questions?